

Lab 02

Intro to git, HWoo, first reading quiz

Today's plan

01 **Intro to git**

Version control, commands

02 **Good practices with git**

03 **HWoo (practice submission)**

Homework/project submission setup +
try on your own

04 **Reading Checkpoint # 1**

Reading: Ten Simple Rules for
Reproducible Computational Research

Intro to git

Building blocks of programming

Terminal

How you will
navigate & run
things



+

Git

How you will save
& track progress



+

Code

The thing you're
actually building

Version control (VCS)

System that records changes to a file or set of files over time.

So ideally, we can place any file in the computer on version control.

Benefits:

- Revert files back to a previous state
- Review changes made over time
- See who last modified something that might be causing a problem
- If something goes wrong or lose files, you can generally recover easily

So what is Git?

Git is **a** version-control system for tracking changes in computer files and coordinating work on those files among multiple people.

It is a Distributed VCS. So Git does not necessarily rely on a central server to store all the versions of a project's files.

Instead, every user makes a *clone* of a project (collection of files) and has the full history of the project on their own hard drive.

This clone has all of the metadata of the original.

git init

Turns a folder into a git repository

Run once, inside the project folder

Creates a hidden .git/ folder that holds all history

Safe to run in an existing folder full of files

Reminder: a git repo = a regular folder, plus that hidden .git/ tracking database inside it.

git status / git add

See what changed, then choose what to save.

git status — shows modified / new / staged files

git add <file> — stage one specific file

git add . — stage everything that changed

Staging lets you save only some of your changes

git commit

Locks in a snapshot of everything staged — your save point.

-m "message" describes what changed and why

Each commit stores: a snapshot + metadata + a link to its parent

git log / git diff

Look back in time: what changed, and when.

git log — full history: hash, author, date, message

git log --oneline — compact one-line-per-commit view

git diff — line-by-line changes not yet staged

Some things to know about Git

- You use it through the terminal, so it has a set of specific commands
- These commands always start with **git** **[command]**
- It usually comes built-in on your computer (Unix, not Windows);
 - How do we check? Open your terminal, and type: **git --version**
 - This means git is fully local
- It doesn't need internet to work
- Git is **not** GitHub and vice versa

Git vs GitHub

Git

a tool

- Software on your machine
- Saves progress **locally**
- Works through commands

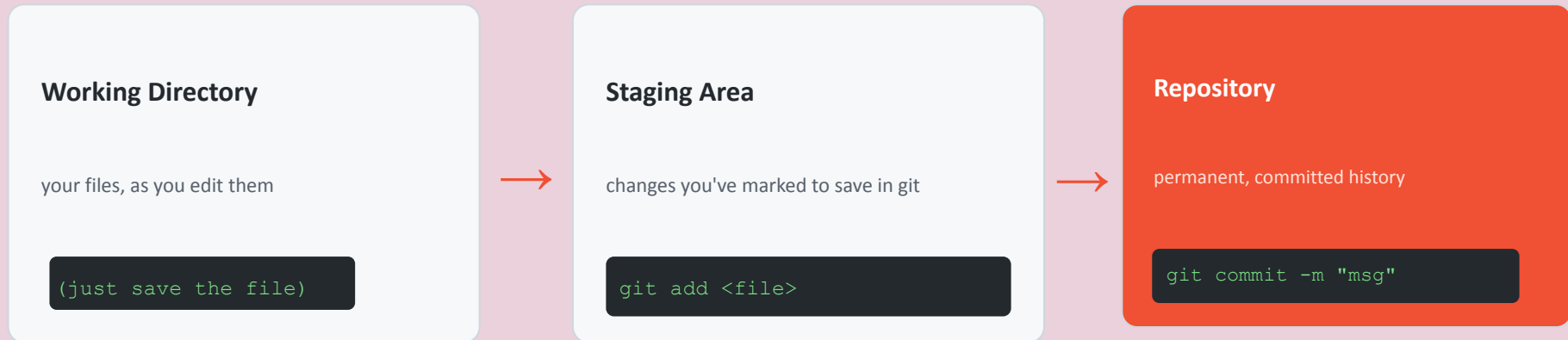
GitHub

a website

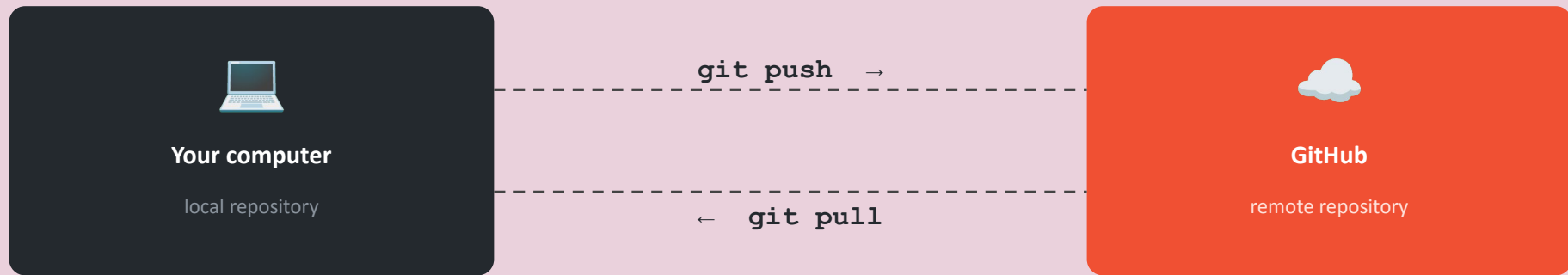
- A place to host your git repos online
- Lets others download/view your code
- Enables real time collaboration
- Other websites similar: GitLab, Bitbucket

List of frequent terms we will be using from now on

- Repo (comes from *repository*) = collection of source code
- Working directory vs staging area vs local repo vs remote repo



Syncing ↔ GitHub



```
$ git remote add origin https://github.com/you/repo.git
$ git push -u origin main      # first push
$ git pull                    # get others' changes
```

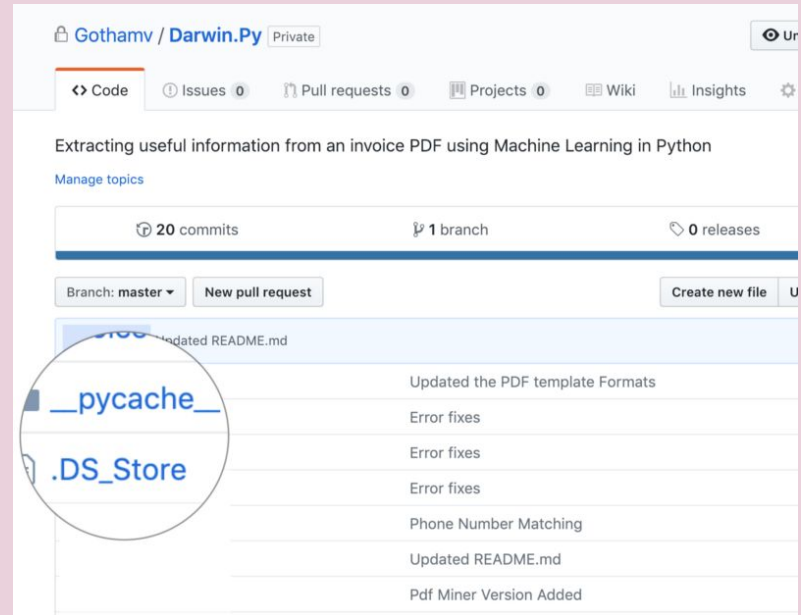
Good practices with Git

- **Commit often:** small, frequent commits are easier to understand and undo.
- **Check status first:** run `git status` before `add/commit`, to know what you're about to save
- **Write clear and explicit commit messages:** "Fix bug" tells no one anything. "Fix off-by-one in sieve_1" does.
- **You don't have to push as often as you commit:** Commit locally as you go; push when a chunk of work is ready to share.

.gitignore

.gitignore is a hidden (text) file, it tells git which files (or patterns) it should ignore.

It's usually used to avoid committing transient files from your working directory that aren't useful to other collaborators, such as compilation products, temporary files IDEs create, etc.



HWoo

1. Join Github Organization

To: Sequoia Andrade <sequoiaandrade@gmail.com>



@fperez has invited you to join the
@UCB-stat-159-f25 organization



@fperez has invited you to join the UCB-stat-159-f25 organization

Hi Sequoia Andrade!

@fperez has invited you to join the @UCB-stat-159-f25 organization on GitHub. Head over to <https://github.com/UCB-stat-159-f25> to check out @UCB-stat-159-f25's profile.

This invitation will expire in 7 days.

[Join @UCB-stat-159-f25](#)

Note: If you get a 404 page, make sure you're signed in as **sequoiarose**.

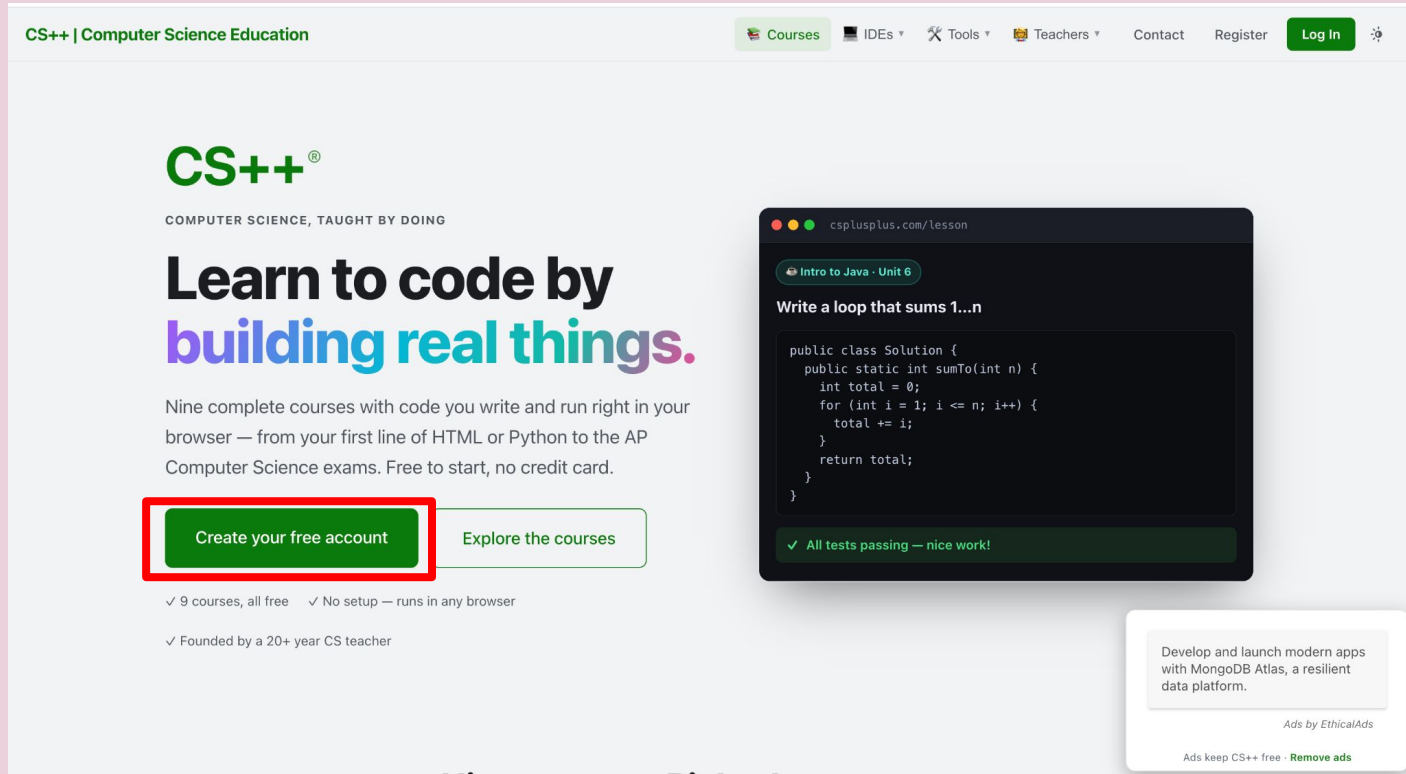
You can also accept the invitation by visiting the organization page directly at <https://github.com/UCB-stat-159-f25>. If @fperez is sending you too many emails, you can [block them](#) or [report them for abuse](#).

Button not working? Paste the following link into your browser:
https://github.com/orgs/UCB-stat-159-f25/invitation?via_email=1

[Opt out of future invitations from this organization](#) •
[Manage your GitHub email preferences](#) • [Terms](#) • [Privacy](#) • [Sign in to GitHub](#)

You're receiving this email because @fperez invited you to an organization on

2. Create an Account on CS ++



The screenshot shows the CS++ website homepage. The navigation bar at the top includes links for Courses, IDEs, Tools, Teachers, Contact, Register, and a Log In button. The main heading reads 'Learn to code by building real things.' Below this, a paragraph states: 'Nine complete courses with code you write and run right in your browser — from your first line of HTML or Python to the AP Computer Science exams. Free to start, no credit card.' Two buttons are present: 'Create your free account' (highlighted with a red box) and 'Explore the courses'. Below the buttons, two bullet points are listed: '✓ 9 courses, all free' and '✓ No setup — runs in any browser'. A second bullet point at the bottom left says '✓ Founded by a 20+ year CS teacher'. On the right side, there is a preview of a lesson window titled 'Intro to Java · Unit 6' with the instruction 'Write a loop that sums 1...n'. The code shown is a Java class 'Solution' with a 'sumTo' method. A green status bar at the bottom of the preview says '✓ All tests passing — nice work!'. At the bottom right, there is an advertisement for MongoDB Atlas with the text 'Develop and launch modern apps with MongoDB Atlas, a resilient data platform.' and a 'Remove ads' link.

CS++[®]
COMPUTER SCIENCE, TAUGHT BY DOING

Learn to code by building real things.

Nine complete courses with code you write and run right in your browser — from your first line of HTML or Python to the AP Computer Science exams. Free to start, no credit card.

[Create your free account](#) [Explore the courses](#)

- ✓ 9 courses, all free
- ✓ No setup — runs in any browser
- ✓ Founded by a 20+ year CS teacher

csplusplus.com/lesson

Intro to Java · Unit 6

Write a loop that sums 1...n

```
public class Solution {  
    public static int sumTo(int n) {  
        int total = 0;  
        for (int i = 1; i <= n; i++) {  
            total += i;  
        }  
        return total;  
    }  
}
```

✓ All tests passing — nice work!

Develop and launch modern apps with MongoDB Atlas, a resilient data platform.

Ads by EthicalAds

Ads keep CS++ free - [Remove ads](#)

3. Join Your Class

CS++ | Computer Science Education

DashboardIDEsCSP ToolsTeachersCOURSE AP CSPSequoia Andrade

Settings

Collapse All

Appearance

Choose how CS++ looks. **System** follows your device's light/dark setting; pick **Light** or **Dark** to override it. Your choice is saved to your account, so it follows you to any computer you sign in on.

**System**
Match my device

**Light**
Always light

**Dark**
Always dark

Join a Classroom

Enter the 6-character code your teacher gave you to join their classroom. Your Code Tracing, Mock Exam, and Debugging Game progress will be shared with your teacher.

Teacher? Looking to add a classroom?
[Go to the Teacher Dashboard →](#)

Code for this lab

Sequoia (11am - 1pm):
A9SSS2

Code for this lab

Larissa (12pm - 2pm):
YRYY2P

Code for this lab

Larissa (4pm - 6pm):
PTAGSV

4. Connect to Github for HW00

CS++ | Computer Science Education

DashboardIDEsCSP ToolsTeachersCOURSE AP CSPSequoia Andrade1

My Classes

Welcome back, Sequoia. Here's what your teachers have queued up for you.

1
CLASS

1
ACTIVE ITEM

0
DUE IN 3 DAYS

0
OVERDUE

Up next: HW 00 · STAT 159/259 — Semester 1 Lab Tues 11a-1pm

Due Aug 28, 11:59 PM (4 days)

STAT 159/259Sequoia Rose Andrade · Semester 1 · Lab Tues 11a-1pm

HW 00

Due Aug 28, 11:59 PM (4 days) ▲

☐ GitHub: hw00

CONNECT GITHUB →

0/1



My Classes

[+ Join Class](#)

Welcome back, Sequoia. Here's what your teachers have queued up for you.

1

CLASS

1

ACTIVE ITEM

0

DUE IN 3 DAYS

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OVERDUE

**Up next:** HW 00 · STAT 159/259 — Semester 1 Lab Tues 11a-1pm

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STAT 159/259

Sequoia Rose Andrade · Semester 1 · Lab Tues 11a-1pm

HW 00

Due Aug 28, 11:59 PM (4 days) ▲

GitHub: hw00 (UCB-stat159-f26/hw00-sequoiarose)

[Set up on your computer →](#) [Wrong account?](#)

not turned in — push a commit

0/1

 Tutor

5. Assignment repo is automatically created for you

The screenshot shows a GitHub repository page for 'hw00-sequoiarose' (Private). The repository was created by 'cs-plus-plus-classroom[bot]' and contains five files: '.gitignore', 'HISTORY.gz', 'README.md', and 'hw00-python-starters.ipynb', all from the initial commit. The README file is open, showing the title 'Stat 159/259 Homework Assignment'. The right sidebar includes an 'About' section with no description, a list of repository features (Readme, Activity, Custom properties, Stars, Watching, Forks, Audit log), and a 'Releases' section with no published releases.

UCB-stat159-f26 / hw00-sequoiarose

Type to search

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

hw00-sequoiarose Private

Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file

Add file

Code

cs-plus-plus-classroom[bot] Initial commit 4526ed3 · 1 minute ago 1 Commit

.gitignore	Initial commit	1 minute ago
HISTORY.gz	Initial commit	1 minute ago
README.md	Initial commit	1 minute ago
hw00-python-starters.ipynb	Initial commit	1 minute ago

README

Stat 159/259 Homework Assignment

About

No description, website, or topics provided.

- Readme
- Activity
- Custom properties
- 0 stars
- 0 watching
- 0 forks
- Audit log

Releases

No releases published

[Create a new release](#)

6. Accept your assignment (by cloning your repo)

Git clone <https://link.to.your.repo.git>

*How to update repos if the template is updated

- Set upstream to template

Git remote add upstream <https://link.to.template>

Git fetch upstream

Git merge upstream/main --allow-unrelated-histories

- Resolve any merge conflicts, commit, then push

Reading Checkpoint #1